ALI MOHAMMADI

Trustworthy AI | Evidence-grounded NLP | Research systems

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Profile

AI researcher and engineering lead working on reliable language-model systems, persistent-memory governance, structured representation, and reproducible evaluation. Research experience spans controlled benchmark design, provenance and revision lineage, ontology-guided representation learning, retrieval systems, and open-source scientific software. Industry work includes enterprise agent platforms, fintech, and vehicle-data systems, with experience carrying projects from research question through implementation and evaluation.

Research interests

Automated fact-checking; evidence-grounded NLP; selective prediction and abstention; temporal and provenance-aware state; trustworthy and agentic AI; long-term memory; knowledge graphs and ontologies; structured evaluation.

Education

M.Eng., Artificial Intelligence and Robotics 2024-2026 (expected)

University of Tehran | GPA 17.85/20 | Adviser: Dr Yadollah Yaghoobzadeh

- Thesis: Keep, Supersede, or Abstain: Interpretable Revision Control for Personalized LLM Memory.
- Selected coursework: Statistical Inference 20/20; Social Network Analysis 20/20; Neural Networks and Deep Learning 19.37/20; Large Language Models 18.4/20.

B.Eng., Computer Engineering (Minor: Software Systems) 2020-2024

University of Tehran | GPA 18.44/20 | Top 5% of cohort

- Thesis: lightweight sequential message authentication for Controller Area Network communication; advised by Professor Mohammad Sayad Haghighi; manuscript in preparation.

Research experience

Interpretable Revision Control for Personalized LLM Memory 2025-present

University of Tehran | M.Eng. thesis | Adviser: Dr Yadollah Yaghoobzadeh

- Designed MGP-6 v2, an author-designed 180-scenario controlled diagnostic separating write decisions, versioned state and lineage, and answer realisation.
- The LLM controller achieved 88.9% write-decision accuracy against 42.2% for naive overwrite; independent annotation of the held-out split reached Cohen's kappa of 0.981.
- Built reproducible controllers, baselines, state and provenance checks, audit ledgers, statistical analyses, and external-framework adapters.

Research Affiliate 2025-present

Cytognosis Foundation | Adviser: Dr Shahin Mohammadi

- Developing ontology-guided representation learning for 1,671 neurobehavioral phenotypes using an EL++-compatible resource curated from HPO, NBO, and ASDPTO.
- Diagnosed leaf-collapse in baseline box embeddings and fine-tuned Gemma 4 E2B-it with LoRA using ontology-graph and box-inclusion objectives; internal evaluation reports parent-retrieval Hits@10 of 0.897 and median rank of 2.
- Built Yar, an evidence-grounded cognitive workspace with source-aware retrieval, persistent structured context, explicit authority and privacy boundaries, abstention, and validation before consequential actions.

PourYa (OuraDataHandler) - Open-source research software 2025

Computational Machinery of Cognition Lab | github.com/CMC-lab/PourYa

- Developed a Python library for ingesting, cleaning, analysing, and visualising Oura Ring data, including API and CSV workflows, change-point detection, tests, documentation, and citation metadata.
- Released version 1.0.0 under the MIT License; the repository includes a software paper, tests, documentation, and citation metadata.

Industry and leadership experience

Innovation Lead and Product Lead, AI and Fintech Platform 2024-present

Injast, Entekhab Holding

- Architected and led a configurable enterprise agent platform for decision support, spanning models, prompts, tools, knowledge sources, integrations, permissions, RAG, speech interfaces, and human handoff.
- Led a 16-person cross-functional team and mentored contributors across research, prototyping, pilot evaluation, and product handover in digital identity, wallet, payments, and lending contexts.

Software Engineer, Machine Learning and Autonomy 2022-2024

Javaneh

- Developed C++ and Python pipelines for motion-data processing, CAN-bus diagnostics, vehicle-network analysis, and predictive-failure modelling for self-driving commercial trucks.

Product Owner / Project Manager, Fintech and Platform Products 2021-2024

DigiPay and Midhco

- Owned product delivery from discovery through launch and directed applied-AI work in fraud detection and credit scoring across payment and lending workflows.

Teaching

Chief Teaching Assistant, Artificial Intelligence 2023-2026

University of Tehran | Dr Yadollah Yaghoobzadeh

- Coordinated the teaching-assistant team for three academic years; mentored students; designed and graded assignments; and led sessions on supervised learning, optimisation, evaluation, and generalisation.

Teaching Assistant, Engineering Probability and Statistics 2022-2023

University of Tehran | Dr Behnam Bahrak

Technical skills

AI and NLP: LLM engineering, agents, retrieval-augmented generation, long-term memory, LoRA fine-tuning, embeddings, selective prediction, abstention, human-in-the-loop systems, statistical evaluation.

Knowledge representation: Knowledge graphs, ontologies (EL++, HPO, NBO, MONDO), box embeddings, graph neural networks, provenance and lineage modelling.

Engineering: Python, C++, SQL, PyTorch, Hugging Face Transformers, FAISS, Ollama, Pydantic, SQLite, Git, Docker, testing, packaging, experiment design.

Awards and languages

- Member, National Organization for Development of Exceptional Talents (SAMPAD), 2018-present.
- Iranian National University Entrance Exam (Mathematics and Physics): rank 221 of approximately 300,000 applicants.
- Best Ideation Award, University of Tehran Summer School, for a subgraph-detection algorithm.
- Languages: Persian (native); English (academic and professional working proficiency).